

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: DRBENCHMARK | Context: EU Pharmaceutical Regulatory NLP — French

Definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Justification

While the benchmark contains broad French pharmaceutical lexical coverage (INNs, dosages, routes, formulations) across multiple datasets (QUAERO-D7, ESSAI-D2, DEFT2021-D3, E3C-D1, CAS-D2), the document genres and registers are predominantly clinical case reports, clinical trial protocols, research abstracts, and spoken prescriptions — all distinct from the formal, QRD-template-constrained EMA regulatory register the deployment targets. QUAERO's EMEA sub-corpus is the only confirmed authentic EMA-document source and covers PILs, not SmPCs (~10 segmented documents) [WEB-3, WEB-4]. Multiple datasets contain non-EMA-jurisdiction content from sub-Saharan Africa, North Africa, Canada, and Middle East (DIAMED-D9, MANTRAGSC-D16, MORFITT-D11, MORFITT-D12, MORFITT-D13, E3C-D15, E3C-D16), introducing construct-irrelevant variance. PxCorpus introduces transcribed-speech artifacts (PXCORPUS-D11, PXCORPUS-D13) that have no analogue in regulatory documents. Brand-name predominance over INNs in PxCorpus (PXCORPUS-D19, PXCORPUS-D20) conflicts with EMA INN-primacy requirements. IC is HIGH priority per the elicitation, and the content register and entity-density profile are systematically misaligned with regulatory prose.

ID	Evidence
WEB-3	QUAERO HuggingFace confirms EMEA = patient leaflet text
WEB-4	QUAERO official corpus page documents 103,056 words across drug leaflets and biomedical titles

Strengths

- Broad French pharmaceutical vocabulary represented across diverse contexts: INNs, dosages, routes, formulations [Q19], (QUAERO-D7, ESSAI-D2, DEFT2021-D3)
- QUAERO EMEA contains authentic EMA leaflet text with authorization headers, full excipient lists with E-numbers, and contraindication statements (QUAERO-D1, QUAERO-D4, QUAERO-D11, QUAERO-D14)
- ESSAI clinical trial protocols contain dense INN + posology vocabulary closer to regulatory text than other clinical sources (ESSAI-D2, ESSAI-D4)
- DEFT2020 STS includes drug-leaflet sentence pairs with named INNs and safety/contraindication content (DEFT2020-D1, DEFT2020-D2, DEFT2020-D7)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: The deployment requires regulatory French in EMA QRD-templated register; the benchmark predominantly contains clinical and research French. The QRD template mandates standardized phrasing 'whenever they are applicable' [WEB-2,

WEB-7], but no DrBenchmark corpus is documented as covering this template-constrained register. Sentence-splitting of EMEA documents [Q37] further removes document-level regulatory context (DEFT2020-D21).

ID	Evidence
WEB-2	EUPATI confirms QRD template mandatory standard statements 'must be used whenever applicable'
WEB-7	EMA confirms SmPC/PIL templates as the regulatory document standard
DEFT2020-D21	En cas de prise le soir, il faut recommander de ne pas s'alter dans les 2 heures suivant celle-ci et de tenir compte du délai d'action (6-8 heures). — Posology instruction decontextualized by sentence splitting; section/drug identity lost

IC-2: EMA/ANSM regulatory culture is well-defined and does not raise sensitive-content concerns; however, the cultural fit issue is reversed — the benchmark contains content from non-EU jurisdictions that dilutes IC validity (DIAMED-D9: Burkina Faso; MORFITT-D11: Saudi Arabia; MORFITT-D12: Egypt; MORFITT-D13: Jordan; MANTRAGSC-D16: Dakar; E3C-D16: Morocco).

ID	Evidence
DIAMED-D9	Communication interventriculaire ischémique: à propos d'un cas observé dans le service de cardiologie du CHU-Yalgado Ouedraogo de Ouagadougou (Burkina Faso) — Sub-Saharan African hospital case; outside EU regulatory jurisdiction and register
MORFITT-D11	Perceptions, attitudes et pratiques vis-à-vis de la recherche chez des médecins internes en formation en Arabie saoudite — Saudi Arabian medical training context; outside EMA/ANSM regulatory scope
MORFITT-D12	Le questionnaire final en arabe (YFAS 2.0-A)...a été rempli par 236 étudiants de médecine égyptiens — Egyptian Arabic-language study; non-French, non-EMA regulatory context
MORFITT-D13	79 patients au total ont été recrutés à Amman (Jordanie) en 2015 — Jordanian clinical context; outside EU regulatory jurisdiction
MANTRAGSC-D16	"Tétanos localisé (à propos de 19 cas observés à Dakar)" — Sub-Saharan African clinical context; outside EMA/ANSM regulatory jurisdiction
E3C-D16	Monsieur F M âgé de 70 ans, originaire du Maroc — North African patient origin; French text but non-EMA/ANSM jurisdictional context

IC-3: The deployment is itself within Western/EU regulatory culture, so the standard 'Western knowledge bias' framing inverts here. The relevant concern is whether the benchmark's content reflects EU-specific regulatory framing — and largely it does not, as much content is research/clinical rather than regulatory (CLISTER-D11, E3C-D11, MORFITT-D8). Brand name vs. INN convention also mismatches EMA INN primacy (PXCORPUS-D19).

ID	Evidence
PXCORPUS-D19	prozac 20 milligrammes 1 gélule matin et soir pendant 1 mois — Brand name (Prozac) used instead of INN (fluoxétine); regulatory mismatch
CLISTER-D11	La patiente a eu, dans le même temps opératoire, une lithotritie balistique du calcul par voie endoscopique permettant de dénuder le DIU puis de l'extraire par une pince à corps étranger. — Clinical case narrative with materiovigilance reference; not regulatory template prose
E3C-D11	Nous rapportons ici le cas d'une patiente de 89 ans aux antécédents d'une diverticulose du sigmoïde — Canonical clinical case reporting register, not regulatory prose
MORFITT-D8	Des rats ont été assignés aléatoirement à 1 de 3 groupes : un groupe témoin (n=15), un groupe (n=15) dans lequel la pancréatite aiguë a été induite au moyen de L-arginine — Animal experimental protocol; research abstract register, not regulatory submission language

IC-4: INSUFFICIENT DOCUMENTATION — no regional regulatory-affairs annotators or reviewers are documented as having validated content for EMA/ANSM register fit. Would need a regulatory-affairs-led review of the QUAERO EMEA subset and any supplementary corpus to confirm representational fit.

IC-5: Content issues harming validity: (a) genre register mismatch (clinical/research vs. regulatory), (b) non-EU-jurisdiction content dilution, (c) brand-over-INN bias in PxCorpus, (d) sentence-splitting context loss for long EMEA documents [Q37], (e) absent SmPC/CTD/EU-RMP source text [WEB-3, WEB-4].

ID	Evidence
WEB-3	QUAERO HuggingFace confirms EMEA = patient leaflet text
WEB-4	QUAERO official corpus page documents 103,056 words across drug leaflets and biomedical titles

QUAERO-D1	"`EMEA / H / C / 122 REFLUDAN Qu' est-ce que Refluidan ? [...] Son principe actif est la lépirudine .'" — EMEA authorization header with brand name and INN — canonical SmPC/leaflet structure
QUAERO-D4	"`Phosphate de sodium , monobasique , monohydraté [...] Polysorbate 80 (E433) Eau pour préparation injectable .'" — Full excipient list with E-numbers in SmPC Section 6 format
PXCORPUS-D19	prozac 20 milligrammes 1 gélule matin et soir pendant 1 mois — Brand name (Prozac) used instead of INN (fluoxétine); regulatory mismatch

Information Gaps

- No published study comparing EMA regulatory French register characteristics to clinical French in NLP corpora (gap_id 5 PARTIALLY RESOLVED)
- No regulatory-affairs annotator review of QUAERO EMEA subset for SmPC/PIL register fit

Requires Expert Verification

- Whether the QUAERO EMEA subset is sufficiently representative of current QRD v10.4 phrasing conventions [WEB-7] for the deployment's needs
- Whether the deployment's supplementary template layer for CTD modules can compensate for the absence of CTD-genre training/evaluation content

Remediation

Gap 1: Non-EU-jurisdiction content (sub-Saharan Africa, North Africa, Middle East, Canada) dilutes IC validity; brand-name predominance in PxCorpus conflicts with EMA INN primacy; sentence-splitting strips long-document context.

Recommendation: When fine-tuning on DrBenchmark, weight or filter to prioritize the QUAERO EMEA, Mantra-GSC EMEA/Patents, and ESSAI subsets that are closest to EU regulatory content. Consider augmenting with deployment-internal SmPC/CTD/RMP text. Develop a long-document handling strategy beyond simple sentence-splitting (e.g., section-aware chunking) for SmPC/CTD inputs.